

Ethernet Redundancy Core

The Ethernet Redundancy Core is a high-performance networking IP designed to provide seamless fault tolerance and zero-loss communication in high-availability Ethernet systems. It implements redundant data paths to ensure continuous operation during link failures, enabling reliable packet delivery for mission-critical and real-time applications on FPGA-based platforms.

The IP core supports hardware-based frame replication, path monitoring, and automatic failover using industry-standard redundancy mechanisms, ensuring uninterrupted data flow without packet loss. Hardware-validated performance, fully synthesizable RTL, and comprehensive verification collateral make the Ethernet Redundancy Core well suited for industrial, aerospace, and high-reliability networking applications.

Introduction

Ethernet redundancy is a hardware-based fault-tolerance technique that ensures uninterrupted data delivery, zero packet loss, and deterministic communication in high-availability Ethernet networks. The Ethernet Redundancy Core implements redundant data paths and rapid failover mechanisms in hardware, forming a critical data-path block for mission-critical and real-time networking applications.

The IP core supports frame replication, duplicate elimination, path supervision, and automatic recovery during link or node failures, ensuring seamless communication without traffic interruption. Advanced hardware-accelerated processing, validated performance, and seamless integration with Ethernet MACs and higher-layer interfaces enable deployment in industrial, aerospace, and safety-critical systems requiring high reliability, low latency, and deterministic operation.

Architecture

The Ethernet Redundancy Core architecture is designed to provide seamless fault tolerance and deterministic Ethernet communication through parallel and redundant data paths. Incoming Ethernet frames from the MAC are processed by the redundancy control logic, where frames are replicated and transmitted simultaneously over multiple network paths.

On the receive side, duplicate frames are detected and eliminated using sequence and supervision mechanisms, ensuring only valid data is forwarded to upper layers. The architecture includes path monitoring, status management, and failover control blocks that continuously supervise link health and enable zero-time recovery in case of link or node failure. Configuration, statistics, and control interfaces allow software supervision, while the fully hardware-based data path ensures low latency, high reliability, and deterministic performance.

Why Choose Us?

 Rapid integration, faster time-to-market

 Clean, verified, documented RTL

 Direct access to experts



Features

- Complete Ethernet Redundancy IP Core providing hardware-based fault tolerance for FPGA-based networking systems.
- Supports 1+1 redundant Ethernet links with hot-standby or load-sharing operation for continuous data delivery.
- Hardware-based frame replication and duplicate elimination ensuring zero packet loss during link or node failures.
- Provides deterministic, low-latency operation suitable for real-time and mission-critical applications.
- Implements industry-standard redundancy mechanisms such as Parallel Redundancy Protocol (PRP) and High-Availability Seamless Redundancy (HSR).
- Compatible with Xilinx and Intel (Altera) FPGA platforms, with standard streaming interfaces for easy integration.
- Single-clock, fully pipelined design enabling straightforward system integration.

Layer-2 Redundancy Capabilities

- Frame replication at ingress and duplicate elimination at egress.
- Continuous path and link supervision with automatic failover and recovery.
- Sequence number insertion and checking for loss and duplication detection.
- Supports 1G and 10G Ethernet operation.
- Transparent to upper-layer protocols and applications.

Management and Integration

- Configurable redundancy modes and supervision parameters.
- Statistics and status reporting for link health and redundancy performance.
- Designed for seamless integration with Ethernet MACs, switches, and higher-layer protocol stacks.

Specifications

S No	Description	Specification
1.	Redundancy	• 1+1 2 user links, the second link is always a hot standby. Load has to be shared by both links, in case of one link failure other link should automatically take over without any loss
2.	Loss	• Zero loss redundancy for 10G and 1G ethernet links
3.	Logic IP Core	• Parallel Redundancy Protocol[PRP] Or High Availability Seamless Redundancy [HSRP] protocol
4.	Support	• Link Aggregation Std IEEE 802.3ad



Specifications Cont...

5.	Compatibility	• ZCU 216 and VPK 180
6.	Integration	• The IP Cores shall be integrated with the ZCU 216 and VPK 180
7.	Documentation	• Shall Submit RTL source code complete project • User guide • Test case generated in questasim • Static code analysis report
8.	Acceptance criteria	• Complete demonstration in the integrated environment with above mentioned speed (10G and 1G Ethernet). • Source code shall be provided to CABS • Industry partner shall carry out all integration & testing activities at CABS. • Complete delivery of IP core integrated with above mentioned boards shall be delivered within 1 year from SO placed

Deliverables

- Basic net-list according to IP features selected
- Example Design for board validation
- Source Code (Optional)
- IP Customization & Integration support (Optional)

Get in touch with us

Delivering Advanced Solutions for
Next-Gen Networking



104, FF, Iris Tech Park, Sector 48
Sohna Road, Gurugram, 122018



+91 124 4048731



sales@qbitlabs.co.in